

Radix-4 NR4SD⁻

non-redundant signed digits of B with an early carry

Step 1 — the number

B is an n -bit two's complement value (n even). The top bit counts negative:

$$B = -2^{n-1}B^{n-1} + \sum_{j=0}^{n-2} 2^j B^j$$

Step 2 — recode the bit pattern in pairs, with a carry

A pair plus its carry in gives a partial sum; a sum of 2 or more is pushed up as a carry, one step earlier than NR4SD⁺:

$$s_j = 2B^{2j+1} + B^{2j} + c_j \quad c_{j+1} = \lceil s_j / 2 \rceil \quad d_j = s_j - 2c_{j+1} \in \{-2, -1, 0, 1\} \quad c_0 = 0$$

d_j depends on the carry, hence on every lower bit: not a local function of its pair.

B^{2j+1}	B^{2j}	c_j	s_j	d_j	c_{j+1}
0	0	0	0	0	0
0	0	1	1	1	0
0	1	0	1	1	0
0	1	1	2	-2	1
1	0	0	2	-2	1
1	0	1	3	-1	1
1	1	0	3	-1	1
1	1	1	4	0	1

Step 3 — the top digit

The pairs count B^{n-1} as $+2^{n-1}$, Step 1 as -2^{n-1} : the difference $-2^n = -4^{n/2}$ has the weight of the last carry, so together they are one extra digit:

$$d_{n/2} = c_{n/2} - B^{n-1} \in \{-1, 0, 1\}$$

Step 4 — the number and the product

$$B = \sum_{j=0}^{n/2} 4^j d_j \quad AB = \sum_{j=0}^{n/2} 4^j d_j A$$

$d_j A \in \{0, A, -A, -2A\}$: the shifted multiple only appears negated, so the shift and the negation always combine. $n/2 + 1$ digits, $n/2 + 1$ partial products. Four digit values in radix four: non-redundant, two control bits, at the price of a carry chain, the extra digit and the $-2A$ multiple.